value. One answer was policy authoring and analysis. These activities are clearly outside the enforcement path, yet they have a significant impact on the quality, accuracy, and maintainability of authorization systems.

Authorization policies must represent organizational intent in a precise, unambiguous, and deterministic manner. Converting that intent into executable policy is often challenging, especially as policy sets grow and interact in complex ways. Even experienced engineers can find it hard to predict the effects of small changes or to explain why a request is allowed or denied.

ACME discovered that AI tools are well-suited for these design-time tasks. When used carefully, they can help draft candidate policies, clarify policy logic in plain language, and identify edge cases that might otherwise be overlooked. None of these places AI in the enforcement path. The policy engine remains the ultimate authority, with AI serving as an assistant to the humans responsible for policy design.

> ***NOTE***
>
> This section explains how ACME uses AI tools to help with policy authoring and analysis at a conceptual level. Appendix B provides a practical, step-by-step walkthrough of this workflow using the ACME Cedar model and an AI-enabled editor. Readers interested in concrete examples, prompts, and policy files can refer to that appendix while reading this section.

17.2.1 Explaining existing behavior

One of the earliest uses of AI at ACME was explaining why a specific authorization request led to a particular outcome. For example, an engineer working on access control for the Customer Collaboration platform might ask:

> *"Why is Kate allowed to view the Q3 plan document?"*